



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201/ COMMON TO ALL BRANCHES (EXCEPT CSBS)

UNIT I – FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS LP 08 – LOGISTIC GROWTH

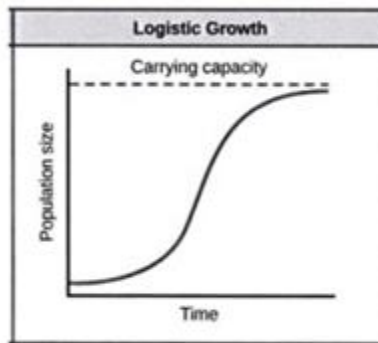
UNIT 1

LOGISTIC GROWTH MODEL

We have seen many examples of exponential population growth based on an equation $\frac{dy}{dt} = ky$, saying intuitively that a larger population produces proportionately more offspring. However, as we have discussed, exponential growth cannot continue forever in a finite ecosystem. There are always limits to growth. One way to model ecological constraints is by adding an extra factor corresponding to the carrying capacity 'M' of the ecosystems:

$$\frac{dP}{dt} = kP(M - P) \quad (\text{or}) \quad \frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

Here 'k' (the continuous growth rate) and M are constants. This model is called a logistic model of growth. As 'x' increases from 0 to M, the factor $1 - \frac{x}{M}$ decreases from 1 to 0, expressing the idea that as the population increases towards carrying capacity, growth becomes increasingly difficult; at carrying capacity growth drops to zero.



The general logistic growth differential equation is given by

$$\frac{dP}{dt} = kP(M - P)$$

We can find the logistic growth model by solving the differential equation,

$$\begin{aligned} \frac{dP}{dt} &= kP(M - P) \\ \frac{dP}{P(M - P)} &= k dt \end{aligned}$$

Integrating on both sides, we get

$$\int \frac{dP}{P(M-P)} = \int k dt$$

We can solve the LHS term by partial fraction method,

Take
$$\frac{1}{P(M-P)} = \frac{A}{P} + \frac{B}{M-P} \text{-----} > (1)$$

$$1 = A(M-P) + BP$$

Let P=0, we get

$$1 = AM + 0$$

$$\Rightarrow A = \frac{1}{M}$$

Let P=M, we get

$$1 = A(0) + BM$$

$$\Rightarrow B = \frac{1}{M}$$

Substitute in eq.(1), we get

$$\frac{1}{P(M-P)} = \frac{1}{M} \frac{1}{P} + \frac{1}{M} \frac{1}{M-P}$$

$$\Rightarrow \int \left(\frac{1}{MP} + \frac{1}{M(M-P)} \right) dP = \int k dt$$

$$\Rightarrow \int \left(\frac{1}{P} + \frac{1}{(M-P)} \right) dP = \int k M dt$$

Integrating on both sides, we get

$$\Rightarrow \ln(P) - \ln(M-P) = kMt + C$$

Multiply by (-1) on both sides, we get

$$\frac{M-P}{P} = e^{-kMt+C_2}$$

$$\frac{M-P}{P} = e^{-kMt_2} \cdot e^{C_2}$$

$$\Rightarrow \frac{M-P}{P} = C_3 e^{-kMt_2}$$

Let $C_3 = A$,

$$M - P = PAe^{-kMt}$$

$$M = P + PAe^{-kMt}$$

$$M = P(1 + Ae^{-kMt})$$

$$\Rightarrow P = \frac{M}{(1 + Ae^{-kMt})}$$

Therefore the solution of the general logistic differential equation $\frac{dP}{dt} = kP(M - P)$ is

$$P = \frac{M}{(1 + Ae^{-kMt})}$$

where M = carrying capacity,

k = relative growth rate, and

A = constant determined by an appropriate initial condition.

NOTE :

The population is growing the fastest when it is half the carrying capacity or when $P = M/2$.

Problems :

1. Given $\frac{dP}{dt} = 0.0008P(700 - P)$, where 't' is years
 - (a) Find the carrying capacity.
 - (b) Find the size of the population when it is growing the fastest.
 - (c) Find the rate at which the population is growing when it is growing the fastest.

Solution:

- (a) Given $\frac{dP}{dt} = 0.0008P(700 - P)$

Here $M = 700$, $k=0.0008$

Carrying capacity $M = 700$

- (b) The size of the population when it is growing the fastest is $P = M/2 = 700/2=350$
- (c) The rate at which the population is growing when it is growing the fastest.

$$\frac{dP}{dt} = 0.0008(350)(700 - 350)$$

$$\frac{dP}{dt} = 98$$

2. A flu epidemic spreads throughout a 3000-student community college at a rate proportional to the number of those already infected and the number of those not yet infected. Let $P(t)$ represent the number of students infected 't' days after the first student gets sick. At time 0, one student is sick. After two days, five students are ill.
- Write a logistic growth model for the spread of the flu.
 - How many students will be sick after ten days?
 - How long will it take until all of the students are infected?

Solution:

- (a) Given $P(0)=1$, $P(2)=5$ & Carrying capacity $M = 3000$
Use $P(2) = 5$ to find k & M ; Use $P(0)=1$ to find A .

We know that the solution of logistic growth model is $P = \frac{M}{1 + Ae^{-kMt}}$.

$$\text{At } t=0, P(0) = \frac{3000}{1 + Ae^0} \Rightarrow 1 = \frac{3000}{1 + A} \Rightarrow 1 + A = 3000 \Rightarrow A = 2999$$

Another formula to find A is $A = \frac{L - P(0)}{P(0)}$

$$\begin{aligned} P(2) &= \frac{3000}{1 + 2999e^{(-k)(3000)(2)}} \Rightarrow 5 = \frac{3000}{1 + 2999e^{(-k)(3000)(2)}} \\ \Rightarrow 1 + 2999e^{(-k)(3000)(2)} &= \frac{3000}{5} \\ 2999e^{(-k)(3000)(2)} &= 600 - 1 = 599 \\ e^{-6000k} &= \frac{599}{2999} = 0.1997 \end{aligned}$$

Taking natural log on both sides (ln), we get

$$k = \frac{\ln(0.1997)}{-6000} \approx 0.000268462$$

Therefore, the logistic growth model for the spread of the flu is $P(t) = \frac{3000}{1 + 2999e^{-0.804t}}$

- (b) Number of students sick after 10 days

$$\text{At } t = 10, \text{ we have } P(10) = \frac{3000}{1 + 2999e^{-0.8054(10)}} = \frac{3000}{1.95317} = 1535.96 \approx 1536 \text{ students}$$

$$(c) \quad 2999.5 = \frac{3000}{1 + 2999e^{-0.8054t}}$$

$$\Rightarrow 1 + 2999e^{-0.8054t} = \frac{3000}{2999.5}$$

$$2999e^{-0.8054t} = \frac{3000}{2999.5} - 1$$

$$e^{-0.8054t} = \frac{\frac{3000}{2999.5} - 1}{2999}$$

Taking ln on both sides, we get

$$t = \frac{\ln \left(\frac{\frac{3000}{2999.5} - 1}{2999} \right)}{-0.8054} \approx 20.742 \approx 21 \text{ days.}$$

3. Solve the initial value problem $\frac{dP}{dt} = 0.0008P(700 - P)$ and $P=10$ when $t=0$.

Solution:

$$\frac{dP}{P(700 - P)} = 0.0008dt$$

Integrate on both sides, we get $\int \frac{dP}{P(700 - P)} = \int 0.0008dt$

Consider the LHS and solve by using partial fraction method

$$\frac{1}{P(700 - P)} = \frac{A}{P} + \frac{B}{700 - P}$$

$$\Rightarrow 1 = A(700 - P) + BP$$

Put $P=0$, $1=A(700 - 0)+B(0)$, which implies $A=1/700$

Put $P=700$, $1=A(0)+700 B$, which implies $B=1/700$

$$\int \frac{dP}{P(700-P)} = \int 0.0008 dt$$

$$\Rightarrow \frac{1}{P(700-P)} = \frac{A}{P} + \frac{B}{(700-P)}$$

$$\int \left(\frac{1}{700P} + \frac{1}{700(700-P)} \right) dP = \int 0.0008 dt$$

$$\int \left(\frac{1}{P} + \frac{1}{(700-P)} \right) dP = \int 700 \times 0.0008 dt$$

$$\int \left(\frac{1}{P} + \frac{1}{(700-P)} \right) dP = \int 0.56 dt$$

$$\ln P - \ln(700-P) = 0.56t + C$$

$$\ln(700-P) - \ln P = -0.56t - C$$

$$\ln(700-P) - \ln P = -0.56t + C_2$$

$$\ln \left(\frac{700-P}{P} \right) = -0.56t + C_2$$

$$\Rightarrow \left(\frac{700-P}{P} \right) = e^{-0.56t + C_2}$$

$$\Rightarrow \left(\frac{700-P}{P} \right) = C_3 e^{-0.56t}$$

P=10, when t=0, we get $\left(\frac{700-10}{10} \right) = C_3 e^{-0.56(0)}$

$$\Rightarrow C_3 = 69$$

$$\Rightarrow \left(\frac{700-P}{P} \right) = 69 e^{-0.56t}$$

$$700 = P + 69P e^{-0.56t}$$

$$700 = P(1 + 69 e^{-0.56t})$$

$$\Rightarrow P = \frac{700}{1 + 69 e^{-0.56t}}$$

Practice Problems :

1. A virus epidemic was formally announced by the city government. According to medical data, it took 5 days for the virus to infect from 3 to 376 people. If the city has 107,524 residents, how many residents are predicted to be infected after 15 days assuming no measures were taken yet by city officials? (Use Logistic model)

2. The population $P(t)$ of bacteria in a petri dish satisfies the logistic differential equation $\frac{dP}{dt} = 0.08P\left(6 - \frac{P}{1000}\right)$ where 't' is measured in hours and the initial population is 200

bacteria. Use the solution to find the population size at $t = 20$ and $t = 100$ hours.

- (i) What is the carrying capacity of the population?
- (ii) What is the population's size when it's growing the fastest?
- (iii) Sketch for different values of 't'.

Discussion Questions

Try to answer the following questions

1. Define logistic growth function.
2. Find the solution of the logistic differential equation.
3. Difference between exponential growth and logistic growth..
4. What is carrying capacity?
5. Sketch the graph of logistic growth function.
6. A lake is stocked with 100 fish. After 3 months, there are 250 fish. A study of the ecology of the lake predicts that lake can support 1000 fish. Find a formula for the number $P(t)$ of fish in the lake 't' months after it has been stocked.
7. A news item is spread by word of mouth to a potential audience of 10000 people. After 't' days $f(t) = \frac{10000}{1 + 50e^{-0.4t}}$ people will have heard the news.
 - (a) Approximately how many people will have heard the news after 7 days?
 - (b) At approximately what rate will the news spread after 14 days?
 - (c) Approximately when will 7000 people have heard the news.
 - (d) When will the news spread at the greatest rate?